1. Let $f(t)$ be a continuous function on $[0,\infty)$. Assume that there exist $A>0$ and $c\in\mathbb{R}$ such that $|f(t)|\le Ae^{ct}$ for t sufficiently large. The *Laplace transform* is an integral of the form

$$Lf(z)=\int_0^\infty f(t)e^{-zt}dt.$$

1) Show that the Laplace transform Lf is holomorphic in the half-plane $\{\mathsf{Re}\, z>c\}$. (c is called the *abscissa of convergence of the integral*.)
2) Assume that f is of class C^1. Show that $Lf'(z)=zLf(z)-f(0)$.

2. Let P be a complex polynomial of degree $n\ge 2$, with n not necessarily distinct zeros $z_1,\cdots,z_n$. Prove the following *Gauss-Lucas theorem*:
For each zero ζ of P', there exist n non-negative real numbers $\lambda_1,\cdots,\lambda_n$ with $\sum_{j=1}^n\lambda_j=1$ such that $\zeta=\sum_{j=1}^n\lambda_j z_j$.
Hint. Note that for $z\in\mathbb{C}\backslash\{z_1,\cdots,z_n\}$,

$$\frac{P'(z)}{P(z)}=\sum_{j=1}^n\frac{1}{z-z_j}=\sum_{j=1}^n\frac{\overline{z-z_j}}{|z-z_j|^2}.$$

3. Let Ω be a domain and $f\in A(\Omega)$. Show that, for any $z_0\in\Omega$ and any open neighborhood U of z_0 in Ω, there exist two points $z_1,z_2\in U$ such that

$$\frac{f(z_2)-f(z_1)}{z_2-z_1}=f'(z_0).$$

Hint. Consider the function $g(z):=f(z)-zf'(z_0)$.

4. 1) Prove the following L^2 version of Riemann's removable singularity theorem. If f has an isolated singularity at z_0 and is L^2-integrable near z_0, then z_0 is a removable singularity for f.
Hint. Let $\sum_{n=-\infty}^\infty a_n(z-z_0)^n$ be the Laurent series expansion of f. Use the L^2 integrability to show that $a_n=0$ for all $n<0$.
2) Give an example to show that the L^2 integrability is optimal, i.e., the theorem does not hold if assuming only L^p integrability for any $p<2$.

5. Let $\Omega=\{z\in\mathbb{C}:\ a<\Im z<b\}$, $-\infty\le a<b\le\infty$. Let $f\in A(\Omega)$ with period 1, i.e. $f(z+1)=f(z)$ for all $z\in\Omega$. Prove that f can be represented as the *complex Fourier series*:

$$f(z)=\sum_{n=-\infty}^\infty a_n e^{i2\pi nz},$$

which is normally convergent on Ω. And the *Fourier coefficients* a_n, $n\in\mathbb{Z}$, are given by the formula

$$a_n=\int_0^1 f(x+iy)e^{-i2\pi n(x+iy)}dx,$$

where $y\in(a,b)$ is arbitrary.
Hint. Consider the map $w=e^{i2\pi z}$ and the function $g(w)=g(e^{i2\pi z})=f(z)$.